

Research Paper Summary

Molecular Mechanisms of Phytochemicals from Chaga Mushroom (*Inonotus obliquus*) Against Colorectal Cancer: Insights from Network Pharmacology, Molecular Docking, and Bioinformatics

Colorectal cancer (CRC) remains one of the leading causes of cancer-related mortality worldwide, with poor prognosis and limited treatment options, particularly in advanced stages. In recent years, natural products have gained increasing attention as potential sources of anticancer agents. One promising candidate is Chaga mushroom (*Inonotus obliquus*), a medicinal mushroom traditionally used in Asia and Eastern Europe that exhibits antioxidant, anti-inflammatory, immunomodulatory, and anticancer properties. However, the molecular mechanisms underlying its therapeutic effects against CRC remain poorly understood.

To address this knowledge gap, this study integrated network pharmacology, molecular docking, and bioinformatics to systematically investigate the molecular mechanisms of Chaga mushroom phytochemicals against CRC, thereby offering a theoretical basis for developing novel CRC treatments.

The study collected Chaga mushroom bioactive compounds from the literature and PubChem database, while colorectal cancer-related targets were obtained from GeneCards, OMIM, and DrugBank. Potential protein targets of the compounds were predicted via SWISSTargetPrediction and standardized using UniProt. Overlapping targets were identified and used to construct a drug–compound–target–disease network with Cytoscape. Protein–protein interaction (PPI) analysis was performed using STRING and Cytoscape to identify hub genes, followed by Gene Ontology (GO) and KEGG pathway enrichment analyses using DAVID. Molecular docking evaluated the binding affinities between key compounds and hub proteins. Furthermore, bioinformatics analyses based on TCGA datasets assessed differential gene expression, immune cell infiltration, and patient survival to validate the biological significance of the identified targets.

This study identified 26 bioactive compounds from Chaga mushroom and 244 overlapping therapeutic targets associated with CRC. The five most influential compounds from network analysis were quercetin, epigallocatechin-3-gallate (EGCG), kaempferol, myricetin, and isorhamnetin. PPI analysis highlighted nine core targets: AKT1, IFNG, MMP9, MMP2, IL6, NFKB1, TNF, CD4, and IL1B. GO and KEGG enrichment analyses revealed that these targets are involved in protein phosphorylation, inflammatory responses, and pathways such as TNF, IL-17, and AGE-RAGE signaling. Molecular docking demonstrated strong binding affinities (mostly below -7 kcal/mol), indicating stable interactions between phytochemicals and target proteins. Bioinformatics analyses identified 7,409 differentially expressed genes, with core genes upregulated in tumor tissues. Immune infiltration analysis revealed substantial differences in immune cell populations, while high expression of IFNG, MMP9, and IL1B was associated with poorer overall survival, highlighting their clinical relevance.

Strengths of this study include the comprehensive integration of network pharmacology, molecular docking, and bioinformatics approaches to systematically elucidate the molecular mechanisms of Chaga mushroom phytochemicals against CRC. This multidisciplinary strategy provides broad and mechanistic insights into its potential anticancer effects. However, the findings are based entirely on computational analyses and require further validation through *in vitro* and *in vivo*

experiments. In addition, the analysis focused only on CRC without comparing effects across other cancer types.

Overall, this paper demonstrates that Chaga mushroom exhibits anticancer activity through a multi-component, multi-target, and multi-pathway mechanism, a defining characteristic of natural product-based therapeutics. Rather than acting on a single molecular target, multiple phytochemicals, including quercetin, EGCG, kaempferol, myricetin, and isorhamnetin, simultaneously interact with multiple proteins and signaling pathways involved in CRC progression.

Methods and Results Illustration

